BG 1st SEMESTER (NEP)
COMPUTER APPLICATIONS (MAJOR)

Session: December 2022 No. of Questions: 16 No. of Pages: 01
Time Allowed: 2:00 Hours Max. Marks: 100 Min. Marks: 40

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

SECTION A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in about 20 words each.

1. Define BIOS.
2. What is the 8-bit one's (1's) complement of binary number 6?
3. What do you mean by open-source software?
4. What is a multi-user operating system?
5. What is the meaning of simplex communication?
6. What is a firewall used for?
7. Name any four smartphone applications used for online education.
8. What is an ISP?

SECTION B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in about 150 words each.

9. What do you mean by data representation? Convert the following into the specified number system:

a) $(31)_{10} = (\ )_{2}$ b) $(11100)_{2} = (\ )_{10}$

c) $(93)_{10} = (\ )_{8}$ d) $(109)_{10} = (\ )_{16}$

OR

Draw a block diagram of a basic computer and explain its working.

10. Tabulate the differences between a compiler and an interpreter.

OR

Using appropriate examples, differentiate between application software and system software.

11. Discuss various modes of communication.

OR

Write short notes on: (i) Router (ii) Switch.

12. What do you mean by E-Governance? Discuss various E-Governance initiatives of the UT of Jammu and Kashmir.

OR

What is Netiquette? Discuss various Internet etiquette rules.

SECTION C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in about 400 words each.

13. Discuss the design and characteristics of memory hierarchy in a computer system.
14. Explain computer language and its various types.
15. Differentiate between LAN, MAN and WAN.
16. Explain the concepts of URL and discuss its various parts. Describe the role of Domain Name Server.